

palmsource™

PalmSource Seminar Series
Build Tomorrow ▸▸▸ Today

Presented By:
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Developer
Architecture
and Disclosure

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Sneak Peek: Palm OS 6

Why this talk?

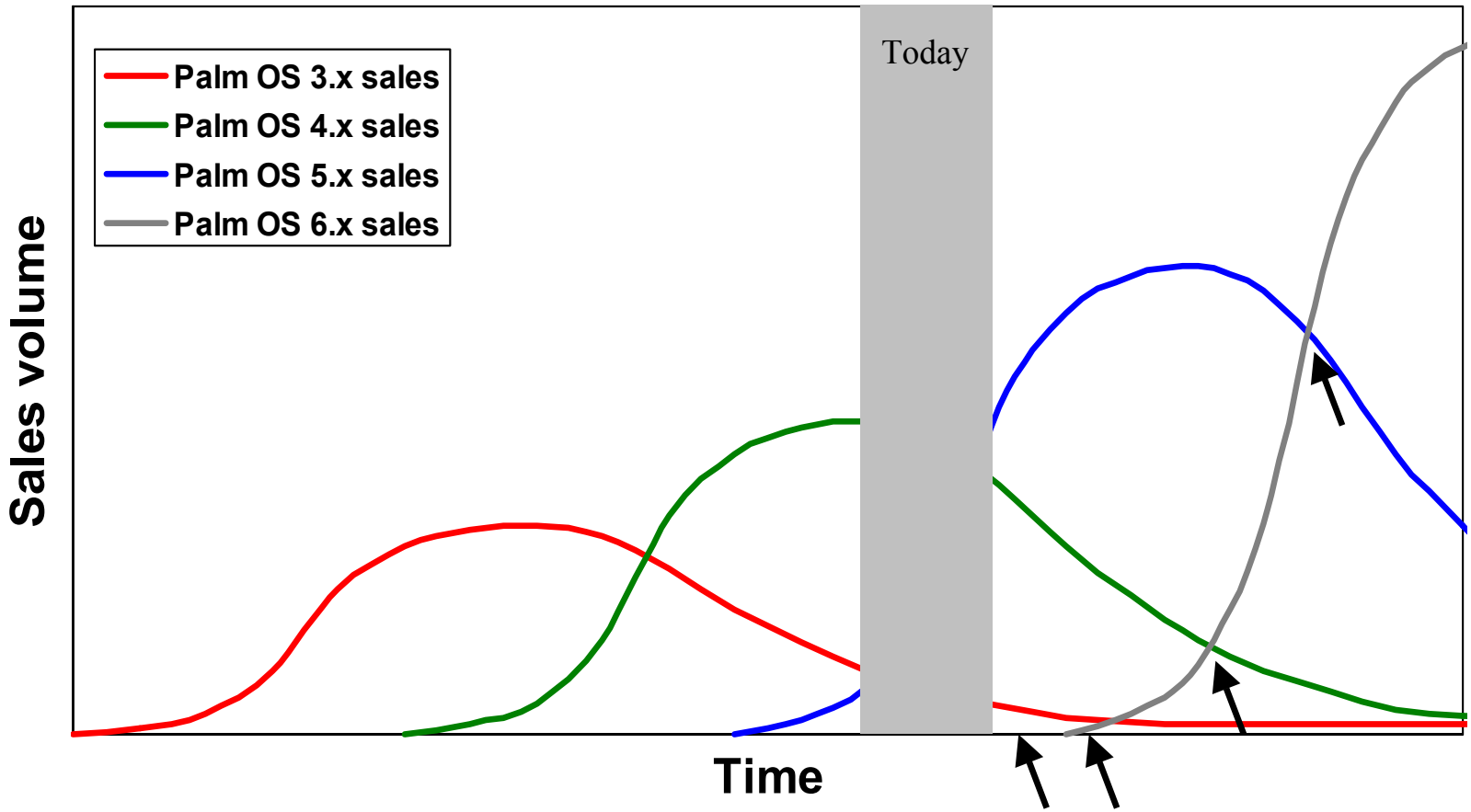
- **You need to plan your future**
- **PalmSource is creating Palm OS 6**

- **Long term focus; talking about the technology as opposed to the applications thereof**
- **Time to prepare for a smooth transition**

(Disclaimer: forward-looking statements abound, but the future cannot be guaranteed!)

Idealized Palm OS sales, by OS release

(Chart is for illustration purposes only; shows neither actual nor projected sales)



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The future affects you today

- **The OS will mature and grow**
  - New features
  - Improvements
  - Things it has grown out of
- **The code you write today has a lifespan**
  - (hopefully a long one!)
- **Features might get easier to create**

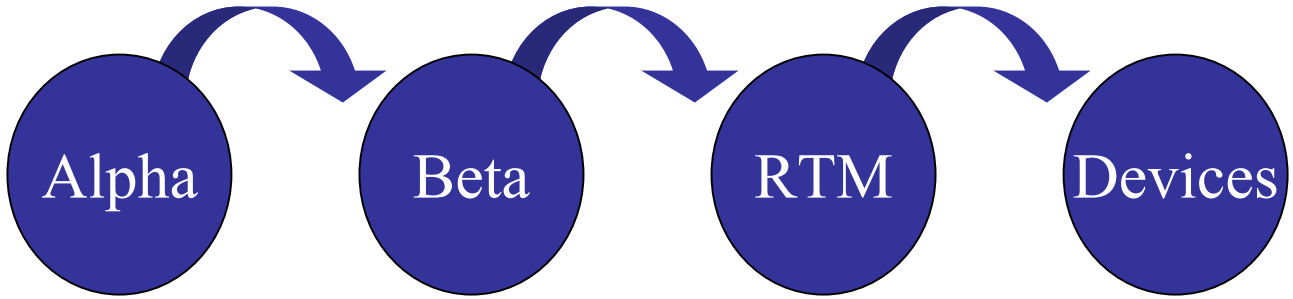
Three effect categories

Every new OS release affects you in 3 ways:

- Existing application compatibility
  - (users upgrading their device)
- New device and OS features
  - (with your old application)
- New APIs, new tools, new SDKs
  - (for your new app releases)

Timeframes

- **Palm OS 6 will soon start its Beta cycle**
- **It is scheduled for handoff to licensees in the fall**
- **Timing of new devices is up to licensees**



Existing Application Compatibility

Compatibility is similar to Palm OS 5.x situation

- PACE runs your existing 68k PRCs
- ARMlets in your 68k PRCs still ideal for speed optimization

Assumptions about OS implementation are dangerous

- The internals are considerably revamped
- Deprecated and unsupported APIs are doomed
- Public APIs are supported

High degrees of compatibility are expected

- (Assuming your applications are well behaved!)

Change is threatening

People want big improvements.

People want nothing to change.

So… nothing is changing,
and everything is changing.

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Changing everything, but nothing

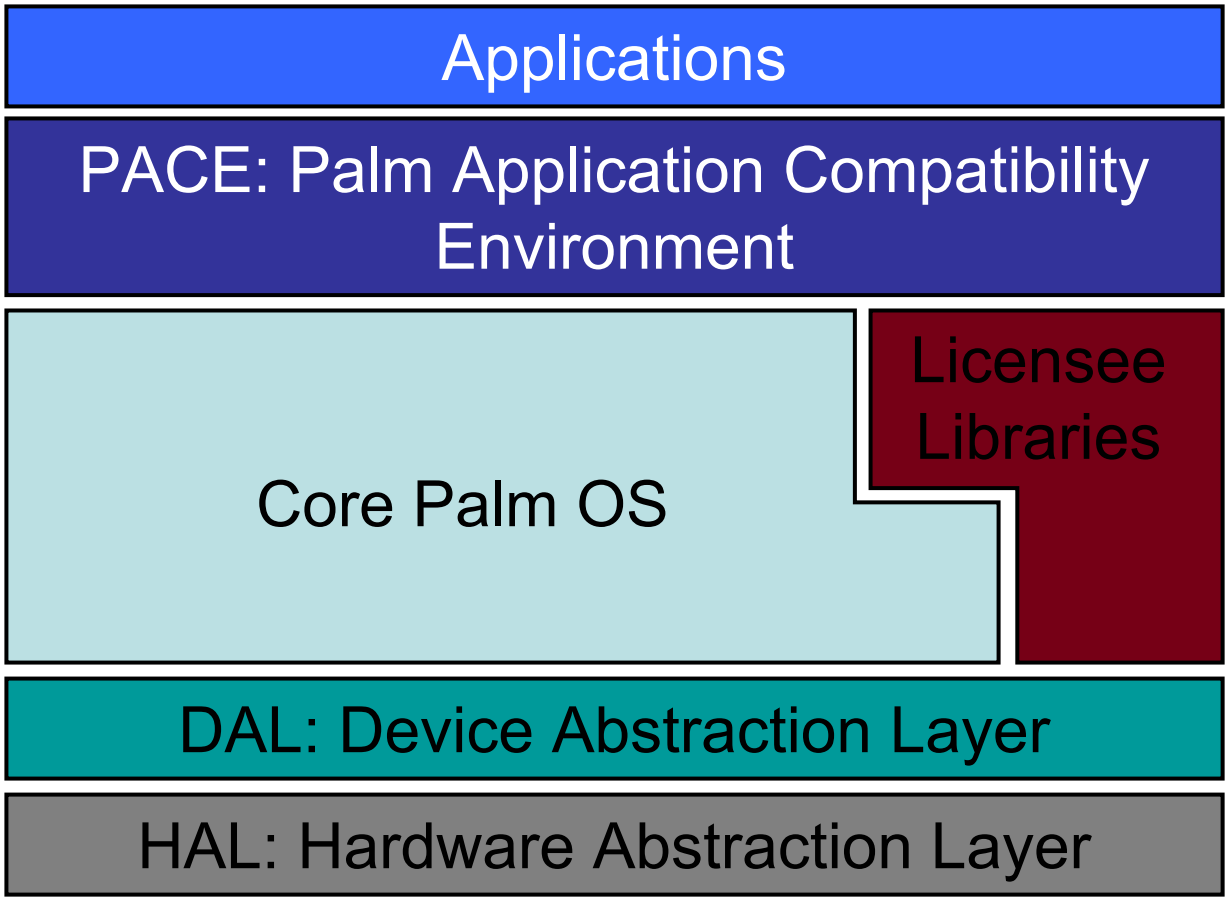
For applications in PACE, the world doesn't change

- (much.)
- ARMlets were designed for compatibility too.
- This is a perfectly valid model to use for years.

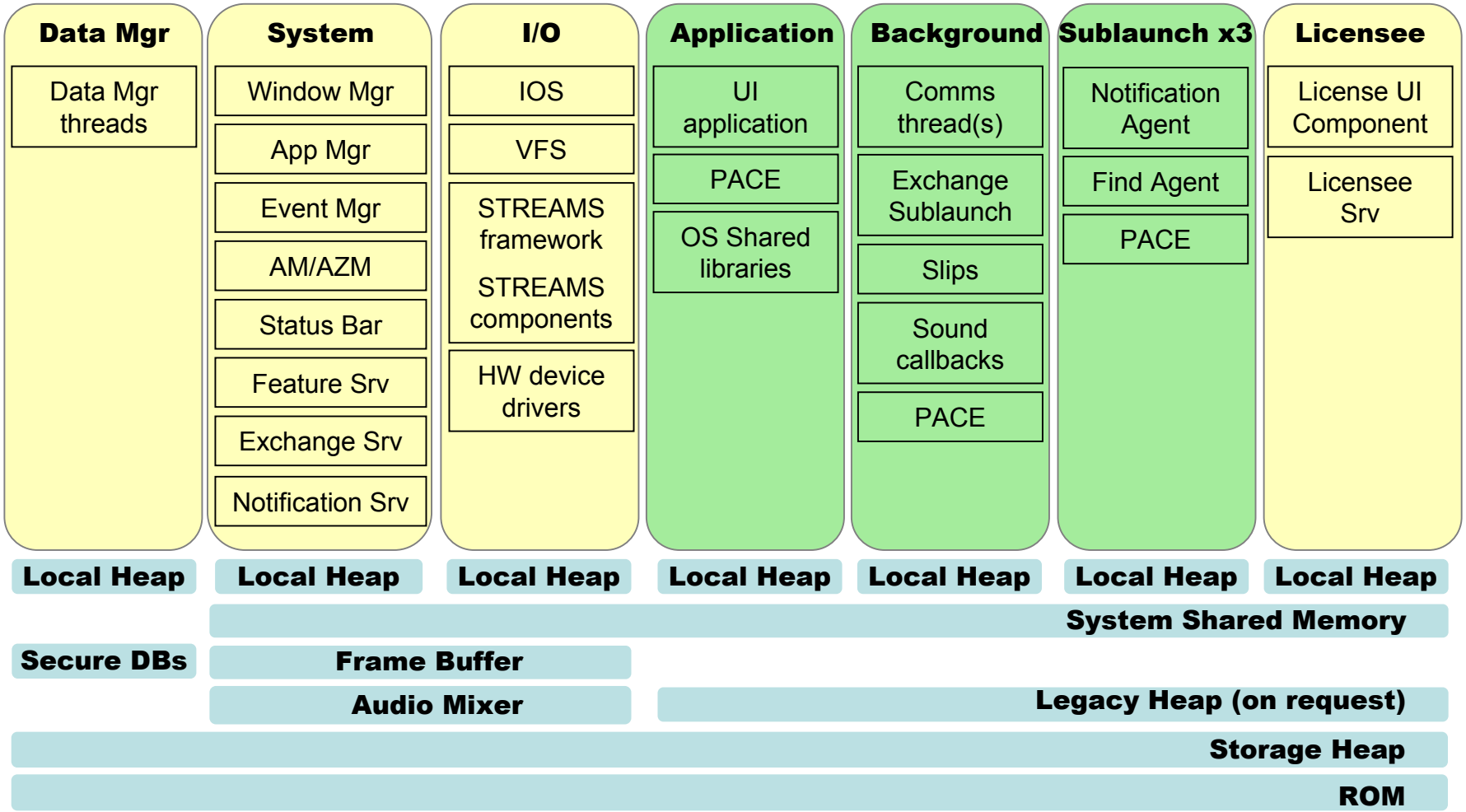
New APIs and capabilities are available to native applications

- This does involve writing new code and tweaking old code.
- Native ARM applications require Palm OS 6.

Block Diagram for 5.x



Palm OS 6.0 Processes and Memory Mapping



Developer-oriented capabilities of the new OS

Protected memory system

- (with your very own dynamic heap)

Process protection

- (for both stability and security)

New runtime model for flexible apps and libraries

Extended database architecture

- for interoperability and extensibility

Threading and Background processes

Communications architecture

Benefits and removing limitations

The top 3 old limitations: gone!

- No 64k limit on records, resources or code segments
- No global variable restrictions
- No 15 category limit

Many new APIs

- Details will be provided

Benefits are ready when your app is ready

Implications of the process model

Some things change for your existing apps

- Not all events go to your event loop.
- The hack-like notifications, patching, etc.
- Code signing & patching

Unless you're extending the system, these shouldn't be noticeable by your application

What will be different

The display

- **Screen sizes, and active input areas**
- **Direct access to screen buffer not allowed**
  - For security reasons, and for new features e.g. acceleration
  - Applications have back buffering for compatibility
- **UI changes**

What will be different (cont.)

The memory

- **Sizes of heaps, the number of heaps, the accessibility of heaps**
- **Endianness and structure packing**
- **New kind of database added: schema db.**
- **PIM databases are schema (thus not present with old names)**

Old memory/database APIs still work as before, but new ones will work better and have more capabilities

What will be different (cont.)

The applications and services

- **Many things move to client-server model**
- **Data and memory sharing (can) work differently**
  - legacy heap, MemPtrSetOwner, etc.
- **Security causes interaction changes**

The Tools

Existing tools, sources and SDKS

- Your compiler, POSE, PalmSim 5.x
- Still work just like before.

New tools and SDKS

- New compiler, new Emulator and new PalmSim
- Visual Studio integration available
- New debugger
- New XML format for UI (on desktop, not on device)
- Sources need to handle endianness, and changes typical of an SDK and OS update, when you choose.

Mixing the Old and the New

Mixing old and new data

- Wire formats.
- Endianness of records & resource data in PDBs
- If you change your DBs, change your conduits

Mixing old and new code

- 68k app with ARMlets
- 68k apps calling ARM shared libraries, sublaunching, etc.
- Can you, and should you, create a “fat” prc

Deciding if and when to go native

- PACE today, PACE tomorrow…

Summary

- **Great things are in your future with Palm OS**
- **The compatibility and transition steps are good for you and your users**
- **We'll keep you well informed**

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Q&A